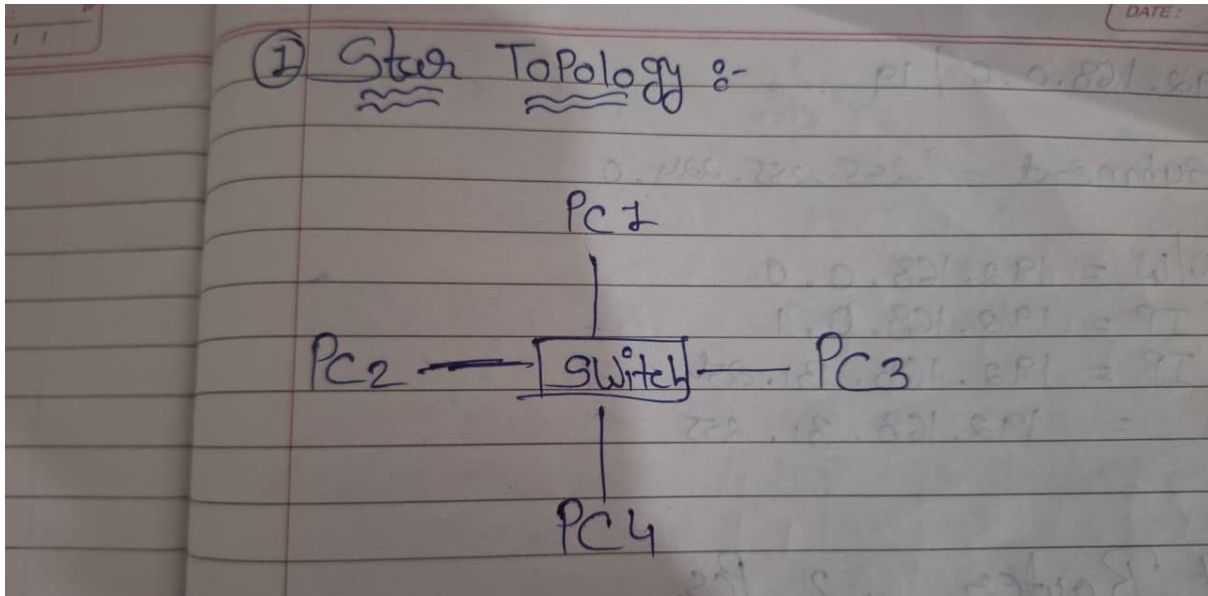


Network Topologies

Task (1):

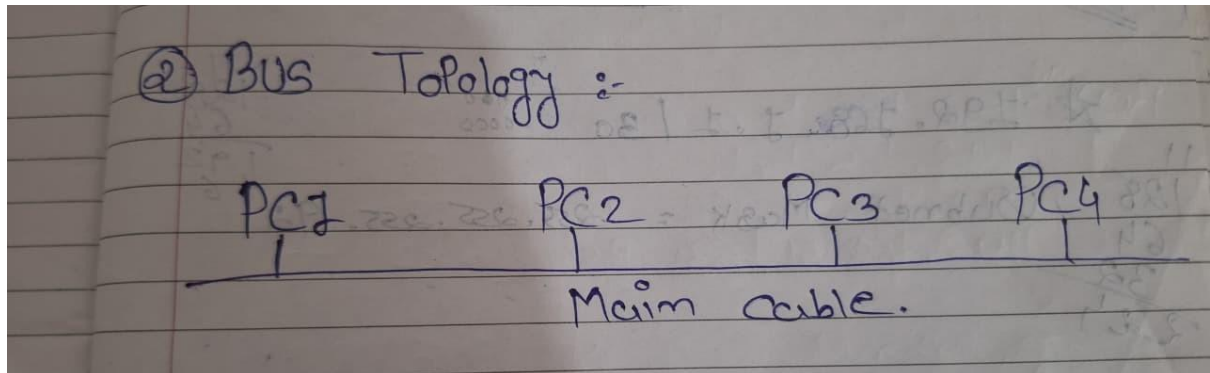
(1)Star Topology:



Real-world example: WhatsApp

Why: A central server manages communication between many users/devices, similar to how a central switch connects devices in a star network.

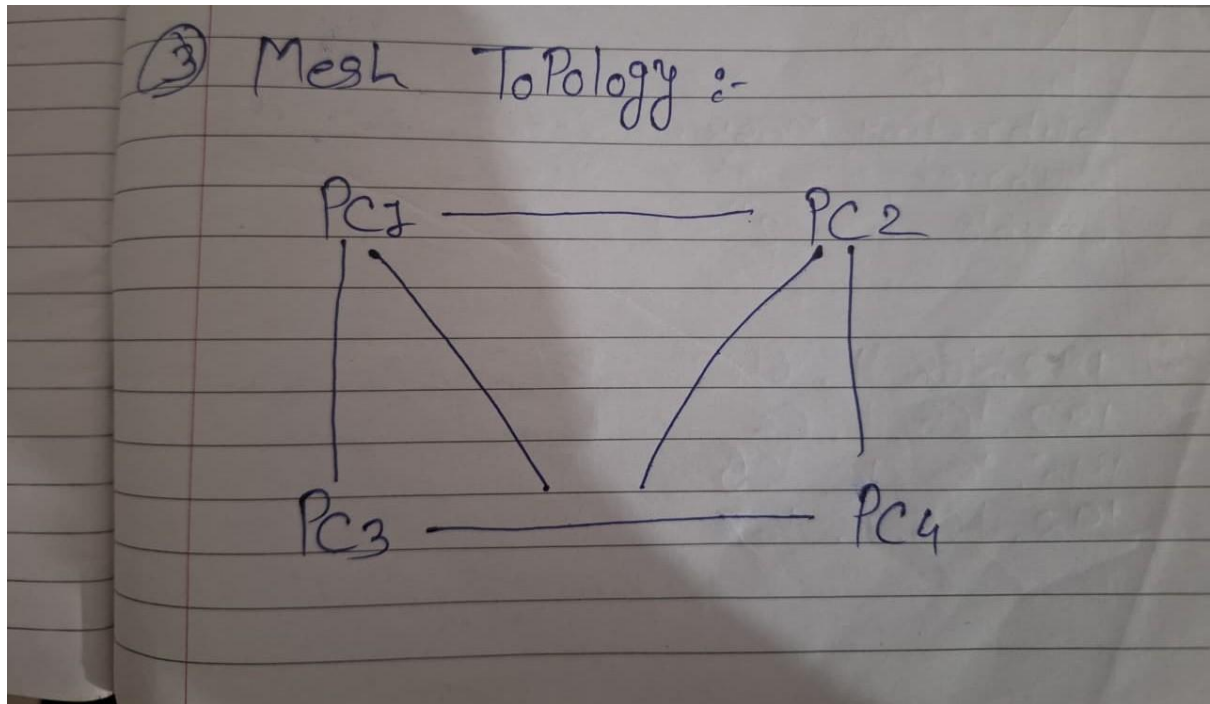
(2) Bus Topology:



Real-world example: Small office/local network

Why: Multiple computers can share one main communication cable, making it simple and inexpensive for small networks.

(3) Mesh Topology:



Real-world example: IPL streaming/CDN network

Why: Multiple network paths can connect servers, so if one connection fails, data can use another path and streaming can continue reliably.

Task (2):

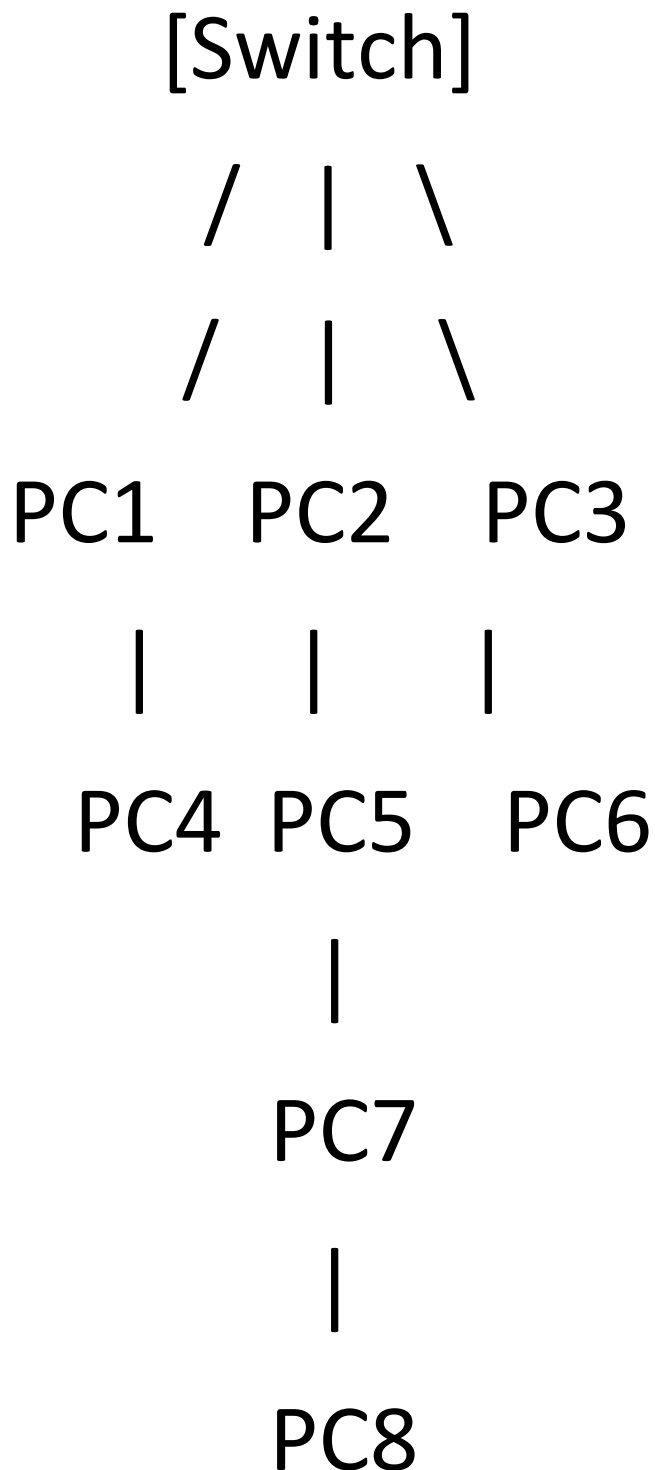
Topology	Pros	Cons	Best Use Case	App Example
Star	1. Easy to install 2. Easy to manage	1. Needs a central switch/hub 2. More cable required.	Offices, schools, and home LANs	Instagram – Central servers manage connections from many users.
Bus	1. Simple design 2. Uses less cable 3. Low cost 4. Easy for small networks	1. Backbone failure stops the network 2. Performance decreases with more devices 3. Difficult to troubleshoot 4. Limited network length	Small temporary or simple networks	Zomato – A simple example of shared communication in a small network environment. Paytm – Networked systems can use organized communication paths for reliable data transfer.
Ring	1. Data travels in an organized way 2. No central device required.	1. One break can affect communication 2. Difficult to troubleshoot.	Specialized or older LAN systems	
Mesh	1. Very reliable 2. Multiple paths for data 3. High fault tolerance 4. Good security and redundancy	1. Expensive 2. Requires many cables/connections 3. Complex to install 4. Difficult to maintain	Data centers, wireless networks, and critical systems	Paytm – Multiple network paths can help keep payment services available if one connection fails.

Task (3):

Suitable Topology: Mesh Topology

- Mesh topology is suitable for a multiplayer game because it provides multiple paths for data to travel.
- It gives fast and reliable communication, so players can continue playing even if one connection fails.
- However, it is more expensive and complex to set up than star or bus topology.

Task (4):



Reason: Star topology is best for a small warehouse because all 8 computers connect to a central switch, providing fast and reliable communication. If one computer or cable fails, the others continue working, and the network is easy to manage and expand.